Supplementary material for "Row-Independent Feature Pre-Screening with Language Models for High-Dimensional Credit Scoring: A Frozen Evaluation on Temporal and Recency-Shift Holdouts"

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Headline subsets

This supplement lists the twelve headline subsets of the main text: for each dataset–backbone case, the features selected by the frozen full-DEV refit of the LLM-assisted family leader and of the classical family leader, in the order the selector produced them ($K = 20$ for logistic regression, $K = 40$ for CatBoost). Each dataset takes two tables, the first giving ranks 1 to 20 of all four subsets and the second ranks 21 to 40 of the two CatBoost subsets: Tables S1 and S2 for Home Credit, Tables S3 and S4 for LendingClub v2, and Tables S5 and S6 for Stability 2024. For pure LLM the order is that of the cached full-DEV ranking; on Stability 2024 one ranking served both backbones, so its logistic-regression subset is the first twenty names of its CatBoost subset. Feature names are those of the engineered universes described in the main text; on Home Credit the three external scores occupy the first three ranks of the pure LLM list. The lists are also provided as `headline_subsets.csv` in the repository.

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Table S1: Home Credit, ranks 1 to 20: full-DEV subsets of the two family leaders under each backbone (primary held-out comparison of the main text), in selector order. The two logistic-regression subsets share 4 of 20 features; the two CatBoost subsets share 16 of 40 (continued in Table S2).

Rank	LR: Stable core + LLM fill (LLM-assisted)	LR: mRMR (classical)	CatBoost: Pure LLM (LLM-assisted)	CatBoost: RFE CatBoost (classical)
1	EXT_SOURCE_2	EXT_SOURCE_3	EXT_SOURCE_3	EXT_SOURCE_2
2	EXT_SOURCE_3	EXT_SOURCE_2	EXT_SOURCE_2	EXT_SOURCE_3
3	PREV_INTEREST_ESTIMATE_MAX	CODE_GENDER	EXT_SOURCE_1	EXT_SOURCE_1
4	EXT_SOURCE_1	CC_CNT_DRAWINGS_ATM_CURRENT_MEAN	DAYS_BIRTH	PREV_INTEREST_ESTIMATE_MAX
5	DAYS_EMPLOYED	POS_LATE_PAYMENT_MEAN	DAYS_EMPLOYED	DAYS_BIRTH
6	BURO_DEBT_CREDIT_DIFF_MIN	DEF_30_CNT_SOCIAL_CIRCLE	OCCUPATION_TYPE	INSTAL_AMT_PAYMENT_SUM
7	INSTAL_IS_LATE_MEAN	PREV_AMT_DOWN_PAYMENT_SUM	NAME_INCOME_TYPE	INSTAL_IS_LATE_MEAN
8	INSTAL_PAYMENT_DIFF_MEAN	FLAG_WORK_PHONE	NAME_EDUCATION_TYPE	DAYS_EMPLOYED
9	PREV_INTEREST_ESTIMATE_VAR	BURO_AMT_CREDIT_SUM_LIMIT_MIN	NAME_FAMILY_STATUS	AMT_CREDIT
10	DAYS_REGISTRATION	CC_AMT_DRAWINGS_ATM_CURRENT_MIN	CODE_GENDER	AMT_ANNUITY
11	POS_SK_DPD_DEF_MEAN	EXT_SOURCE_1	AMT_CREDIT	AMT_GOODS_PRICE
12	BURO_DEBT_CREDIT_DIFF_MEAN	PREV_RATE_INTEREST_PRIMARY_VAR	AMT_GOODS_PRICE	INSTAL_DAYS_DIFF_MAX
13	PREV_INTEREST_ESTIMATE_MEAN	WEEKDAY_APPR_PROCESS_START	AMT_ANNUITY	PREV_DAYS_FIRST_DUE_SUM
14	CC_CNT_DRAWINGS_ATM_CURRENT_MEAN	PREV_DAYS_FIRST_DRAWING_VAR	BURO_DEBT_RATIO_MAX	CODE_GENDER
15	DAYS_BIRTH	COMMONAREA_AVG	BURO_DEBT_RATIO_MEAN	BURO_DEBT_RATIO_SUM
16	AMT_CREDIT	FLAG_CONT_MOBILE	BURO_DAYS_CREDIT_MEAN	BURO_DEBT_RATIO_MAX
17	AMT_GOODS_PRICE	FLAG_DOCUMENT_3	BURO_DAYS_CREDIT_UPDATE_MEAN	INSTAL_DAYS_INSTALMENT_VAR
18	AMT_ANNUITY	PREV_NFLAG_LAST_APPL_IN_DAY_MEAN	BURO_DAYS_CREDIT_ENDDATE_MEAN	PREV_APPLICATION_DIFF_MAX
19	BURO_DEBT_RATIO_MAX	AMT_REQ_CREDIT_BUREAU_QRT	BURO_DAYS_CREDIT_MAX	PREV_DAYS_LAST_DUE_1ST_VERSION_MAX
20	BURO_DEBT_RATIO_MEAN	NAME_HOUSING_TYPE	BURO_DAYS_CREDIT_MIN	BURO_DAYS_CREDIT_ENDDATE_MAX

Table S2: Home Credit, ranks 21 to 40 of the two CatBoost subsets of Table S1.

Rank	CatBoost: Pure LLM (LLM-assisted)	CatBoost: RFE CatBoost (classical)
21	BURO_DAYS_ENDDATE_FACT_MEAN	NAME_EDUCATION_TYPE
22	BURO_DEBT_CREDIT_DIFF_MEAN	BURO_DEBT_CREDIT_DIFF_SUM
23	BURO_DEBT_RATIO_VAR	INSTAL_PAYMENT_RATIO_MEAN
24	BURO_AMT_CREDIT_SUM_DEBT_MEAN	BURO_AMT_CREDIT_SUM_MAX
25	BURO_AMT_CREDIT_SUM_DEBT_SUM	CC_LIMIT_USE_MEAN
26	BURO_AMT_CREDIT_SUM_LIMIT_MEAN	PREV_INTEREST_ESTIMATE_SUM
27	BURO_AMT_CREDIT_SUM_LIMIT_SUM	INSTAL_AMT_PAYMENT_MAX
28	BURO_AMT_CREDIT_SUM_DEBT_VAR	BURO_DAYS_CREDIT_MEAN
29	BURO_DAYS_CREDIT_UPDATE_MAX	DAYS_REGISTRATION
30	BURO_DAYS_CREDIT_ENDDATE_MAX	CC_CNT_DRAWINGS_CURRENT_VAR
31	BURO_DAYS_CREDIT_UPDATE_VAR	REGION_POPULATION_RELATIVE
32	BURO_DAYS_CREDIT_VAR	POS_TOTAL_INSTALMENT_PROG_MEAN
33	BURO_CREDIT_DURATION_MEAN	PREV_DAYS_LAST_DUE_MAX
34	BURO_CREDIT_DURATION_MAX	CC_CNT_DRAWINGS_ATM_CURRENT_MEAN
35	BURO_CREDIT_DURATION_VAR	DAYS_ID_PUBLISH
36	BURO_AMT_ANNUITY_SUM	BURO_DEBT_RATIO_MEAN
37	BURO_CNT_CREDIT_PROLONG_MEAN	INSTAL_PAYMENT_DIFF_MAX
38	BURO_CNT_CREDIT_PROLONG_MAX	OWN_CAR_AGE
39	BURO_CNT_CREDIT_PROLONG_SUM	NAME_FAMILY_STATUS
40	INSTAL_PAYMENT_RATIO_MEAN	BURO_DEBT_CREDIT_DIFF_MIN

Table S3: LendingClub v2, ranks 1 to 20: full-DEV subsets of the two family leaders under each backbone (primary held-out comparison of the main text), in selector order. The two logistic-regression subsets share 4 of 20 features; the two CatBoost subsets share 21 of 40 (continued in Table S4).

Rank	LR: Pure LLM (LLM-assisted)	LR: IV then Boruta (classical)	CatBoost: LLM then mRMR (LLM-assisted)	CatBoost: IV then Boruta (classical)
1	fico_mean	term_home_ownership	term_60 months	term_home_ownership
2	fico_band	term_verification_status	acc_open_past_24mths_share	term_verification_status
3	loan_to_income	term	dti	term
4	dti	term_months	fico_mean	term_months
5	annual_inc	term_x_loan_to_income	term_36 months	term_x_loan_to_income
6	loan_amnt	fico_band_x_term	term_home_ownership_60 months__RENT	fico_band_x_term
7	term	loan_amnt_to_income	term_x_loan_to_income	loan_amnt_to_income
8	home_ownership	loan_to_income	fico_band_x_term_near_prime__60 months	loan_to_income
9	verification_status	loan_to_income_x_fico	acc_open_past_24mths_per_total_acc	loan_to_income_x_fico
10	purpose	fico_mean	term_home_ownership_36 months__MORTGAGE	fico_mean
11	total_bc_limit	fico_midpoint_scaled	fico_range_high	fico_midpoint_scaled
12	bc_open_to_buy	fico_range_high	term_verification_status_60 months__Source Verified	fico_range_high
13	bc_util	fico_range_low	acc_open_past_24mths_per_credit_history_year	fico_range_low
14	total_rev_hi_lim	fico_band	term_verification_status_60 months__Verified	fico_band
15	revol_util	total_bc_limit_to_loan_amnt	fico_range_low	total_bc_limit_to_loan_amnt
16	num_actv_rev_tl	tot_hi_cred_lim_to_loan_amnt	loan_to_income	tot_hi_cred_lim_to_loan_amnt
17	num_rev_tl_bal_gt_0	bc_open_to_buy_to_loan_amnt	log_avg_cur_bal	bc_open_to_buy_to_loan_amnt
18	acc_open_past_24mths	acc_open_past_24mths_per_credit_history_year	fico_midpoint_scaled	acc_open_past_24mths_per_credit_history_year
19	num_tl_op_past_12m	acc_open_past_24mths_per_total_acc	loan_amnt_to_income	acc_open_past_24mths_per_total_acc
20	credit_history_years	acc_open_past_24mths_share	acc_open_past_24mths	acc_open_past_24mths_share

Table S4: LendingClub v2, ranks 21 to 40 of the two CatBoost subsets of Table S3.

Rank	CatBoost: LLM then mRMR (LLM-assisted)	CatBoost: IV then Boruta (classical)
21	sqrt_avg_cur_bal	total_rev_hi_lim_to_loan_amnt
22	dti_x_bc_util	acc_open_past_24mths
23	total_bc_limit_to_loan_amnt	log_acc_open_past_24mths
24	log_acc_open_past_24mths	avg_cur_bal_to_loan_amnt
25	mort_acc_per_open_acc	acc_open_past_24mths_band
26	fico_band_x_term_prime__60 months	dti_x_bc_util
27	num_tl_op_past_12m_per_credit_history_year	dti
28	term_home_ownership_60 months__MORTGAGE	prime_fico_flag
29	term_verification_status_36 months__Not Verified	dti_x_revol_util
30	num_tl_op_past_12m_per_total_acc	dti_band
31	dti_x_revol_util	tot_cur_bal_to_loan_amnt
32	avg_cur_bal	num_tl_op_past_12m_per_credit_history_year
33	loan_to_income_x_fico	loan_to_total_credit_limit_v2
34	total_bc_limit_per_bc_trade	loan_to_total_limit
35	balance_to_high_credit_limit	acc_open_past_24mths_per_open_acc
36	verification_group_x_loan_to_income_band_verified__extreme	balance_to_high_credit_limit
37	bc_open_to_buy_to_loan_amnt	log_num_tl_op_past_12m
38	dti_band_high	num_tl_op_past_12m
39	fico_band_near_prime	num_tl_op_past_12m_per_total_acc
40	mths_since_recent_inq	num_tl_op_past_12m_share

Table S5: Stability 2024, ranks 1 to 20: full-DEV subsets of the two family leaders under each backbone (primary held-out comparison of the main text), in selector order. The two logistic-regression subsets share 1 of 20 features; the two CatBoost subsets share 4 of 40 (continued in Table S6).

Rank	LR: Pure LLM (LLM-assisted)	LR: RFE CatBoost (classical)	CatBoost: Pure LLM (LLM-assisted)	CatBoost: CatBoost SHAP (classical)
1	d0__static__maxdpdlast24m_143P	d1__credit_bureau_a__dpdmaxdateyear_596T__mean	d0__static__maxdpdlast24m_143P	d1__person__sex_738L__lexical_mode
2	d0__static__maxdpdlast12m_727P	d1__credit_bureau_a__numberofoverdueinstlmax_1039L__mean	d0__static__maxdpdlast12m_727P	d0__static__mobilephncnt_593L
3	d0__static__maxdpdlast9m_1059P	d0__static__price_1097A	d0__static__maxdpdlast9m_1059P	d0__static__isbidproduct_1095L
4	d0__static__maxdpdlast6m_474P	d1__credit_bureau_a__dateofcredstart_739D__mean	d0__static__maxdpdlast6m_474P	d1__person__incometype_1044T__first_by_num_group1
5	d0__static__maxdpdlast3m_392P	d1__credit_bureau_a__numberofoverdueinstlmax_1151L__mean	d0__static__maxdpdlast3m_392P	d0__static__avgdpdtolclosure24_3658938P
6	d0__static__maxdpdtolerance_374P	d0__static__lastrejectdate_50D	d0__static__maxdpdtolerance_374P	d1__person__sex_738L__first_by_num_group1
7	d0__static__maxdbddpdtollast12m_3658940P	d0__static__maxdbddpdtollast12m_3658940P	d0__static__maxdbddpdtollast12m_3658940P	d0__static__pmtnum_254L
8	d0__static__maxdbddpdlast1m_3658939P	d1__applprev__maxdpdtolerance_577P__mean	d0__static__maxdbddpdlast1m_3658939P	d1__credit_bureau_a__dpdmaxdateyear_596T__mean
9	d0__static__avgdbddpdlast24m_3658932P	d0__static_cb__pmtssum_45A	d0__static__avgdbddpdlast24m_3658932P	d0__static__annuity_780A
10	d0__static__avgdpdtolclosure24_3658938P	d1__credit_bureau_a__residualamount_856A__mean	d0__static__avgdpdtolclosure24_3658938P	d1__person__education_927M__first_by_num_group1
11	d0__static__actualdpdtolerance_344P	d0__static__pmtnum_254L	d0__static__actualdpdtolerance_344P	d0__static_cb__pmtssum_45A
12	d0__static__numinstlswithdpd10_728L	d1__credit_bureau_a__totalamount_6A__sum	d0__static__numinstlswithdpd10_728L	d0__static__applicationscnt_867L
13	d0__static__numinstlswithdpd5_4187116L	d1__person__birth_259D__max	d0__static__numinstlswithdpd5_4187116L	d0__static_cb__days90_310L
14	d0__static__numinstlswithoutdpd_562L	d1__person__incometype_1044T__first_by_num_group1	d0__static__numinstlswithoutdpd_562L	d1__applprev__education_1138M_last_by_num_group1
15	d0__static__numinstpaidlate1d_3546852L	d0__static__numinstlallpaidearly3d_817L	d0__static__numinstpaidlate1d_3546852L	d1__person__birth_259D__mean
16	d0__static__numinstregularpaid_973L	d1__credit_bureau_a__overdueamountmax2_14A__mean	d0__static__numinstregularpaid_973L	d1__tax_registry_a__amount_4527230A__sum
17	d0__static__numinsttopaygr_769L	d1__tax_registry_a__amount_4527230A__sum	d0__static__numinsttopaygr_769L	d1__tax_registry_a__row_count
18	d0__static__numinstunpaidmax_3546851L	d0__static_cb__days90_310L	d0__static__numinstunpaidmax_3546851L	d1__applprev__maxdpdtolerance_577P__mean
19	d0__static__pctinstlsallpaidlat10d_839L	d1__credit_bureau_a__totalamount_996A__max	d0__static__pctinstlsallpaidlat10d_839L	d1__credit_bureau_a__financialinstitution_591M__first_by_num_group1
20	d0__static__pctinstlsallpaidlate1d_3546856L	d1__credit_bureau_a__overdueamountmax_35A__mean	d0__static__pctinstlsallpaidlate1d_3546856L	d1__credit_bureau_a__refreshdate_3813885D__min

Table S6: Stability 2024, ranks 21 to 40 of the two CatBoost subsets of Table S5.

Rank	CatBoost: Pure LLM (LLM-assisted)	CatBoost: CatBoost SHAP (classical)
21	d0__static__pctinstlsallpaidlate4d_3546849L	d1__tax_registry_c__row_count
22	d0__static__pctinstlsallpaidlate6d_3546844L	d0__static__lastrejectdate_50D
23	d0__static__pctinstlsallpaidearl3d_427L	d1__credit_bureau_a__dateofcredstart_739D__min
24	d0__static__amtinstpaidbefduel24m_4187115A	d1__applprev__status_219L__lexical_mode
25	d0__static__avginstallast24m_3658937A	d1__person__incometype_1044T__lexical_mode
26	d0__static__maxinstallast24m_3658928A	d1__credit_bureau_a__dateofcredstart_739D__mean
27	d0__static__maxoutstandbalancel12m_4187113A	d0__static__lastrejectreason_759M
28	d0__static__sumoutstandtotal_3546847A	d0__static__eir_270L
29	d0__static__totaldebt_9A	d0__static__lastdelinqdate_224D
30	d0__static__currdebt_22A	d1__credit_bureau_a__overdueamountmaxdateyear_2T__mean
31	d0__static__currdebtcredtyperange_828A	d0__static__cntpmts24_3658933L
32	d0__static__disbursedcredamount_1113A	d1__credit_bureau_a__dpdmax_139P__first_by_num_group1
33	d0__static__credamount_770A	d0__static__maxdbddpdtollast12m_3658940P
34	d0__static__totalsettled_863A	d1__credit_bureau_a__dpdmax_139P__mean
35	d0__static__downpmt_116A	d1__credit_bureau_a__dpdmax_757P__mean
36	d0__static__annuity_780A	d0__static__price_1097A
37	d0__static__monthsannuity_845L	d0__static__disbursedcredamount_1113A
38	d0__static__maxdebt4_972A	d1__credit_bureau_a__numberofoverdueinstlmaxdat_148D__max
39	d0__static__numactivecreds_622L	d1__person__birth_259D__first_by_num_group1
40	d0__static__numactiverelcontr_750L	d1__credit_bureau_a__overdueamountmax2date_1142D__max